In this assessment, students will learn about using Ansible to automate installation and other repetitive tasks. The first 3 exercises are interactive labs without the need of installing any SW in your VM. The remaining exercises use Ansible in the local VM.

1. Ansible lab 1

Automate the installation of a MySQL db.

https://www.redhat.com/en/interactive-labs/use-system-roles-install-microsoft-sql-server

Note: the last update on the interactive lab produced some errors:

```
TASK [microsoft.sql.server : Link the deprecated mssql_ad_sql_user_name fact] ***************************************
skipping: [localhost]

TASK [microsoft.sql.server : Verify that the user accepts EULA variables] ******************************************
fatal: [localhost]: FAILED! => {
    "assertion": "mssql_accept_microsoft_sql_server_standard_eula | bool",
    "changed": false,
    "evaluated_to": false,
    "msg": [
        "You must accept EULA by setting the following variables to true:",
        "mssql_accept_microsoft_odbc_driver_17_for_sql_server_eula",
        "mssql_accept_microsoft_cli_utilities_for_sql_server_eula",
        "mssql_accept_microsoft_sql_server_standard_eula"
    ]
}

PLAY RECAP ********************************************************************************************************
localhost                  : ok=4    changed=0    unreachable=0    failed=1    skipped=7    rescued=0    ignored=0

root@rhel:~# vi site.yml
```

Try to fix it changing the vars (or creating new vars) in the playbook.

2. Ansible Lab 2

Automate the configuration of firewall rules.

https://www.redhat.com/en/interactive-labs/configure-firewalls-red-hat-enterprise-linux-system-roles

3. Ansible Lab 3

Automate the system configuration of a RH operating system.

https://www.redhat.com/en/interactive-labs/build-standard-operating-environment-system-roles

4. Ansible Installation on your VM

Install Ansible on your Ubuntu machine:

```
sudo apt-add-repository ppa:ansible/ansible
sudo apt update
sudo apt install ansible
```

The result should be similar to this one:

```
root@guest-VirtualBox:/var/lib/dpkg# ansible --version
ansible 2.9.20
  config file = /etc/ansible/ansible.cfg
  configured module search path = [u'/root/.ansible/plugins/modules', u'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python2.7/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 2.7.12 (default, Mar  1 2021, 11:38:31) [GCC 5.4.0 20160609]
```

Additionally, ssh and daemon sshd should be up and running:

$ which ssh /usr/bin/ssh

$ which sshd /usr/bin/sshd

In case the above commands return error install the ssh:

$ sudo apt-get install ssh

Note: The following exercises have the solution here

5. Get remote host description and characteristics.

With ansible you can check the environment characteristics of the remote hosts (e.g. CPU, OS, network, etc). Try to create an inventory and playbook that gather information about a remote machine. In case you don't have access to a remote machine you can connect to your localhost.

To solve this exercise the following link might be useful:

https://docs.ansible.com/ansible/latest/user_guide/playbooks_vars_facts.html
https://docs.ansible.com/ansible/latest/user_guide/intro_inventory.html

5.1 – How can you change the playbook / inventory to gather information of multiple machines?

5.2 - Change the playbook to add two new tasks:

- Get and print the “Uptime” of the remote machine
- Get the file /var/log/syslog from remote machine

Note1: To get the uptime you can execute the module shell. To print the result, you can use the debug command:

https://docs.ansible.com/ansible/2.4/shell_module.html
https://docs.ansible.com/ansible/2.4/debug_module.html

Note2: To get the file you can use the fetch module:

https://docs.ansible.com/ansible/2.3/fetch_module.html

6. Playbook to install Docker

Create a playbook that installs docker in a machine (or group of machines).

Note: In the first class was mentioned that to install Docker you should do:

$sudo apt-get install curl

$curl -sSL https://get.docker.com/ | sh

$sudo usermod -aG docker guest

7. Running a container through Ansible.

Run a container through Ansible that has the name nginx_<machine name> and that has the port 8080 open.
You must also check that the container is operational in the port 8080 (you can use the “wait_for” module). You can also check that the default page returns with success.

8. Ansible UI

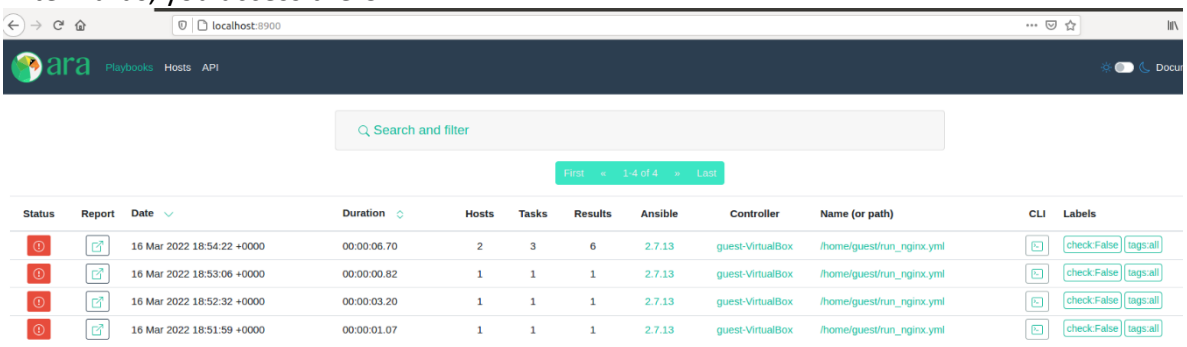
There are two open-source projects for UI:

- Ara - https://github.com/ansible-community/ara
- Ansible AWX - https://github.com/ansible/awx

In this exercise you can try to install and use Ara (or Ansible AWX)
To install Ara you can run the following commands:

```
#with docker from the image on quay.io:
docker run --name api-server --detach --tty \
--volume ~/.ara/server:/opt/ara:z -p 8000:8000 \
quay.io/recordsansible/ara-api:latest
# Install Ansible and ARA (without API server dependencies) for the current user
python3 -m pip install --user ansible ara
# Configure Ansible to use the ARA callback plugin
export ANSIBLE_CALLBACK_PLUGINS="$(python3 -m ara.setup.callback_plugins)"
# Set up the ARA callback to know where the API server is located
export ARA_API_CLIENT="http"
export ARA_API_SERVER="http://127.0.0.1:8000"
# Run an Ansible playbook
ansible-playbook playbook.yaml
# Use the CLI to see recorded playbooks
ara playbook list
```

Afterwards, you access the UI:



The following links help to solve this exercise:

https://docs.ansible.com/ansible/2.6/modules/docker_container_module.html#docker-container-module
https://docs.ansible.com/ansible/latest/collections/ansible/builtin/uri_module.html
https://docs.ansible.com/ansible/2.9/modules/wait_for_module.html

9. Explore Ansible demos

The link https://github.com/ansible/product-demos/tree/main/linux has a lot of demos that show the multiple possibilities related to Ansible. Try to walk through these demos.